

FSM <-> SAP Job Incentives

Integration Mapping Document

Pixelcare Field Service Management | SAP Business One Add-On

Overview of the Data Flow

The FSM portal integrates with the SAP Business One Job Incentives add-on in two directions:

1. FSM PUSHES job, schedule, and technician data INTO SAP so the add-on can compute incentives.
2. FSM FETCHES computed incentive results FROM SAP stored procedures and displays them in the portal.

```

FSM Portal  -> [Push to SAP]  ->  SAP Tables: OCLG, SCL5, @API_JOB_SCHEDULE
                                |
                                SAP Add-On computes incentives
                                Saves results to @JOB_INCENTIVES
                                |
FSM Portal  <- [Fetch from SAP] <-  Stored Procedures (_SPPXC_Job_Incentive_Data, etc.)
  
```

PART 1 -- FSM -> SAP (What You Need to Push)

The SAP add-on reads from 3 key SAP tables. You must populate all 3.

1A. @API_JOB_SCHEDULE -- Job-to-Technician Link

This is the most critical table. Every SAP stored procedure joins through here to determine which technician owns which job.

SAP Column	SAP Type	FSM Table	FSM Column	Notes
U_JobNo	NVARCHAR	jobs	job_number	FSM job number -- must match exactly
U_JobTech	NVARCHAR	technicians	sap_tech_code (!)	SAP tech code e.g. 'A1EdsonBeh' -- NEW field needed
U_CallID	INT	jobs	sap_activity_id	SAP Service Call ID -- already stored in FSM

NOTE: REQUIRED FSM schema addition: Add sap_tech_code varchar to the technicians table. This is the SAP-side identifier like 'A1EdsonBeh', 'C4Yoga'. Without it, the join in every stored procedure breaks.

1B. OCLG -- SAP Activity / Service Call Log (Job Schedule Date)

OCLG.Recontact is the DATE FILTER in all 3 stored procedures. This date determines which month/year a job belongs to.

SAP Column	FSM Table	FSM Column	Notes
ClgCode (PK)	jobs	sap_activity_id	Already synced; used as the join key throughout
Recontact	job_schedule	jsdate	Job schedule date -- MUST be set when creating the SAP activity. This is the month/year filter.

1C. SCL5 -- Service Call Lines (Job Income & Invoice/CM Link)

Line-level records under an OCLG activity. The add-on reads income amounts and document numbers from here.

SAP Column	FSM Table	FSM Column	Notes
ClgID	jobs	sap_activity_id	Links this line to the OCLG activity
U_API_JobNumber	jobs	job_number	FSM job number -- must be written to this custom field
U_InvNumber	jobs	payment_qr_inv_number	AR Invoice doc number from SAP
U_CMNumber	jobs	sap_cm_number (!)	Credit Memo doc number -- NEW field needed on jobs
U_JobIncome	jobs	sap_job_income (!)	Incentive income per job line -- NEW field needed
U_JobStatus	technician_jobs	assignment_status	Push 'I' (Invoiced) when COMPLETED + invoice created in SAP

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U_CMStatus	jobs	sap_cm_status (!)	Push 'WCM' / 'WOCM' -- NEW field needed on jobs
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NOTE: REQUIRED FSM schema additions on the jobs table:

- sap_cm_number varchar (credit memo document number)
- sap_cm_status varchar ('WCM' = With Credit Memo, 'WOCM' = Without Credit Memo)
- sap_job_income numeric (incentive income amount for this job)

PART 2 -- SAP -> FSM (What to Fetch Back)

2A. _SPPXC_Job_Incentive_Data -- Summary View

Called with: (FilterType, Year, Quarter, Month)

Returns one row per technician + one Grand Total row. Use this for the incentives summary/dashboard screen.

FilterType: 'M' = Monthly (only Month 1 columns populated), 'Q' = Quarterly (3 months shown).

Quarter mapping: Q1 = Jan/Feb/Mar, Q2 = Apr/May/June, Q3 = Jul/Aug/Sep, Q4 = Oct/Nov/Dec.

SAP Column	Type	Source Table	Source Column	Description	FSM Display Label	Join / Notes
RowType	string	Computed	--	'D' = tech row, 'G' = grand total	--	Use to style grand total row differently
Tech	string	@API_JOB_SCHEDULE	U_JobTech	SAP technician code	--	Join to technicians.sap_tech_code -> full_name
Year	string	Parameter	@Year	Selected year	Year	
Month1	int	OCLG	MONTH(Recontact)	Month number for column 1	(convert to name)	1=Jan, 2=Feb, etc.
Inc1	decimal	SCL5	U_JobIncome (SUM)	Incentive income -- Month 1	Income (Month 1)	Sum of all job income lines
Exp1	decimal	@JOB_INCENTIVES	U_Expense (SUM)	Expense -- Month 1	Expense (Month 1)	Manually entered in SAP add-on
IncD1	decimal	@JOB_INCENTIVES	U_IncomePerDollar (SUM)	Income per dollar -- Month 1	Income/Dollar (M1)	= Inc1 / Exp1
Work1	decimal	@JOB_INCENTIVES	U_WorkingHrs (SUM)	Working hours -- Month 1	Working Hrs (M1)	Manually entered in SAP add-on
IncH1	decimal	@JOB_INCENTIVES	U_IncomePerHour (SUM)	Income per hour -- Month 1	Income/Hour (M1)	= Inc1 / Work1
Month2	int	OCLG	MONTH(Recontact)	Month number column 2 (Q only)	(convert to name)	Quarterly view only
Inc2	decimal	SCL5	U_JobIncome (SUM)	Incentive income -- Month 2	Income (Month 2)	
Exp2	decimal	@JOB_INCENTIVES	U_Expense (SUM)	Expense -- Month 2	Expense (Month 2)	
IncD2	decimal	@JOB_INCENTIVES	U_IncomePerDollar (SUM)	Income/Dollar -- Month 2	Income/Dollar (M2)	
Work2	decimal	@JOB_INCENTIVES	U_WorkingHrs (SUM)	Working Hrs -- Month 2	Working Hrs (M2)	
IncH2	decimal	@JOB_INCENTIVES	U_IncomePerHour (SUM)	Income/Hour -- Month 2	Income/Hour (M2)	
Month3	int	OCLG	MONTH(Recontact)	Month number column 3 (Q only)	(convert to name)	Quarterly view only
Inc3	decimal	SCL5	U_JobIncome (SUM)	Incentive income -- Month 3	Income (Month 3)	
Exp3	decimal	@JOB_INCENTIVES	U_Expense (SUM)	Expense -- Month 3	Expense (Month 3)	
IncD3	decimal	@JOB_INCENTIVES	U_IncomePerDollar (SUM)	Income/Dollar -- Month 3	Income/Dollar (M3)	
Work3	decimal	@JOB_INCENTIVES	U_WorkingHrs (SUM)	Working Hrs -- Month 3	Working Hrs (M3)	

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IncH3	decimal	@JOB_INCENTIVES	U_IncomePerHour (SUM)	Income/Hour -- Month 3	Income/Hour (M3)	
Inc	decimal	Computed	SUM(Inc1+Inc2+Inc3)	Total income (all months shown)	TOTAL Income	
Exp	decimal	Computed	SUM(Exp1+Exp2+Exp3)	Total expense	TOTAL Expense	
Work	decimal	Computed	SUM(Work1+Work2+Work3)	Total working hours	TOTAL Working Hrs	
IncD	decimal	Computed	SUM(IncD1+IncD2+IncD3)	Total income/dollar	TOTAL Income/Dollar	
IncH	decimal	Computed	SUM(IncH1+IncH2+IncH3)	Total income/hour	TOTAL Income/Hour	

2B. _SPPXC_Technician_Data -- Technician Detail View

Called with: (FilterType, Year, Quarter, Month, Tech, FilterCategory, DocType, DocNumFrom, DocNumTo, DateFrom, DateTo)
Returns one row per invoice/credit memo for a specific technician. Use this for the drill-down detail screen.

FilterCategory values: 'NF' = date filter only, 'WODT' = doc number range, 'WODN' = doc type, 'ALL' = combined.

SAP Column	Type	Description	FSM Display Label	Join / Notes
Job Technician	string	SAP technician code	--	Join to technicians.sap_tech_code -> full_name
Document Type	string	'Invoice' or 'Credit Memo'	Document Type	
Document Date	date	Job schedule date (OCLG.Recontact)	Job Date	
Document Entry	int	SAP internal doc PK (OINV/ORIN)	(internal)	Use for SAP deep-links if needed
Document Number	int	AR Invoice or CM number (user-facing)	Invoice/CM No.	Join to jobs.payment_qr_inv_number or jobs.sap_cm_number
BP Code	string	Business Partner code	Customer Code	Join to customer.customer_code
BP Name	string	Business Partner company name	Customer Name	
Document Amount	decimal	Total document value (OINV.DocTotal)	Document Total	
Incentive Amount	decimal	Sum of SCL5.U_JobIncome for this doc	Incentive Amount	

2C. _SPPXC_DocNum_Data -- Document Number Filter Dropdown

Called with: (FilterType, Year, Quarter, Month, Tech)

Returns all distinct invoice and credit memo numbers for a technician in the period. Used to populate a filter dropdown.

SAP Column	Description	FSM Usage
Document Number	Invoice or CM document number	Dropdown option value
Document Type	'Invoice' or 'Credit Memo'	Dropdown option label prefix

PART 3 -- FSM Schema Changes Required

3A. New Column on technicians Table

```
ALTER TABLE public.technicians
  ADD COLUMN sap_tech_code character varying;
-- e.g. 'A1EdsonBeh', 'C4Yoga'
-- Must match exactly what is stored in @API_JOB_SCHEDULE.U_JobTech in SAP
```

3B. New Columns on jobs Table

```
ALTER TABLE public.jobs
  ADD COLUMN sap_cm_number character varying, -- credit memo doc number from SAP
  ADD COLUMN sap_cm_status character varying, -- 'WCM' = With Credit Memo, 'WOCM' = Without
  ADD COLUMN sap_job_income numeric DEFAULT 0; -- incentive income amount for this job
```

3C. Optional: Cache Tables for Incentive Results

If you want to avoid calling SAP every page load, cache the stored procedure results in Supabase.

```
-- Summary cache (one row per technician per month)
CREATE TABLE public.job_incentive_results (
  id                uuid    NOT NULL DEFAULT uuid_generate_v4(),
  technician_id     uuid    NOT NULL,
  sap_tech_code     varchar NOT NULL,
  year              smallint NOT NULL,
  month             smallint NOT NULL, -- 1-12
  income            numeric DEFAULT 0,
  expense           numeric DEFAULT 0,
  working_hrs       numeric DEFAULT 0,
  income_per_dollar numeric DEFAULT 0,
  income_per_hour   numeric DEFAULT 0,
  fetched_from_sap_at timestamp with time zone DEFAULT now(),
  created_at        timestamp with time zone DEFAULT now(),
  updated_at        timestamp with time zone DEFAULT now(),
  CONSTRAINT job_incentive_results_pkey PRIMARY KEY (id),
  CONSTRAINT job_incentive_results_technician_id_fkey
    FOREIGN KEY (technician_id) REFERENCES public.technicians(id)
);

-- Detail cache (one row per invoice/CM per technician)
CREATE TABLE public.job_incentive_detail_results (
  id                uuid    NOT NULL DEFAULT uuid_generate_v4(),
  job_incentive_result_id uuid NOT NULL,
  technician_id     uuid    NOT NULL,
  document_type     varchar, -- 'Invoice' or 'Credit Memo'
  document_date     date,
  document_entry    varchar,
  document_number   varchar,
  bp_code           varchar,
  bp_name           varchar,
  document_amount   numeric DEFAULT 0,
  incentive_amount  numeric DEFAULT 0,
  fetched_from_sap_at timestamp with time zone DEFAULT now(),
  CONSTRAINT job_incentive_detail_results_pkey PRIMARY KEY (id),
  CONSTRAINT fk_result FOREIGN KEY (job_incentive_result_id)
    REFERENCES public.job_incentive_results(id),
  CONSTRAINT fk_tech FOREIGN KEY (technician_id)
    REFERENCES public.technicians(id)
);
```

PART 4 -- Integration Flow Summary

```
PUSH (FSM -> SAP)
jobs.job_number      -> SCL5.U_API_JobNumber
                    -> @API_JOB_SCHEDULE.U_JobNo
jobs.sap_activity_id -> SCL5.ClgID
                    -> @API_JOB_SCHEDULE.U_CallID
technicians.sap_tech_code -> @API_JOB_SCHEDULE.U_JobTech
job_schedule.jsdate  -> OCLG.Recontact (set when creating SAP activity)
jobs.payment_qr_inv_number -> SCL5.U_InvNumber
jobs.sap_cm_number   -> SCL5.U_CMNumber
jobs.sap_job_income  -> SCL5.U_JobIncome
COMPLETED + invoiced -> SCL5.U_JobStatus = 'I'
has credit memo      -> SCL5.U_CMStatus = 'WCM'
```

```
FETCH (SAP -> FSM)
EXEC _SPPXC_Job_Incentive_Data(FilterType, Year, Quarter, Month)
  -> Summary grid: income/expense/hours per technician per month
  -> Join: Tech -> technicians.sap_tech_code -> technicians.full_name

EXEC _SPPXC_Technician_Data(FilterType, Year, Quarter, Month, Tech, ...)
  -> Detail grid: per-document breakdown for one technician
  -> Join: BP Code -> customer.customer_code
  -> Join: Document Number -> jobs.payment_qr_inv_number / jobs.sap_cm_number

EXEC _SPPXC_DocNum_Data(FilterType, Year, Quarter, Month, Tech)
  -> Filter dropdown values only (doc numbers for the period)
```

Key Dependency

The single field that unlocks everything is `technicians.sap_tech_code`. Without it, the FSM cannot tell SAP which technician owns a job, and all stored procedures will return NULL technician names across all results.